

A Gentle Introduction to Mean Field Games

Optimal transport, variational problems on measures, and forward–backward PDEs

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Abstract

Mean field games describe Nash equilibria in populations containing very many weakly interacting decision makers. Their basic mathematical object is a forward–backward PDE system: a Hamilton–Jacobi–Bellman equation propagates the value of a representative agent backward in time, while a continuity or Fokker–Planck equation propagates the population law forward. This note gives a gentle introduction using the notation and viewpoint of *Optimal Transport for Machine Learners*. The emphasis is on the links with transport of probability measures, Benamou–Brenier actions, first variations of functionals, potential games, Wasserstein geometry, and minimizing movements. It also stresses an important boundary: a generic mean field game is a Nash equilibrium problem, not a Wasserstein gradient flow. The JKO scheme and the global variational formulation of a potential game are closely related but encode different time structures. After deriving the MFG system, we discuss monotonicity, uniqueness, finite-player limits, master equations, examples, and the main numerical strategies.

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1 Why mean field games?

Mean field games (MFGs) were introduced independently by Lasry and Lions (2006a,b, 2007) and by Huang et al. (2006) to describe differential games with a very large number of statistically similar agents. The motivating compromise is simple. A game with N players has an N -body state and, in feedback form, an N -body system of Hamilton–Jacobi equations. Direct computation quickly becomes impossible. Yet replacing the players by one centrally controlled fluid changes the model: selfish agents seek a Nash equilibrium, not the minimizer of a social cost. The mean field limit keeps the strategic character of the game while compressing the other $N - 1$ players into their empirical distribution.

The resulting state variable is a probability measure $\alpha_t \in \mathcal{P}(X)$. This makes MFGs naturally close to optimal transport, mean-field particle systems, and Wasserstein gradient flows. The same continuity and Fokker–Planck equations occur in all these theories, but their interpretations differ:

- in optimal transport, a velocity is chosen to move one prescribed measure to another with minimal kinetic action;
- in a Wasserstein gradient flow, the velocity follows the steepest descent of one functional of the current measure;
- in mean field control, a planner chooses the population dynamics to minimize one collective objective;
- in a mean field game, every infinitesimal agent optimizes against a population curve that must, at equilibrium, be generated by the agents’ optimal responses.

This last fixed-point requirement is the source of the characteristic forward–backward coupling. The population law determines the value function; the value function determines an optimal feedback; and that feedback must reproduce the population law.

Scope. We work mostly on the flat torus $X = \mathbb{T}^d$, which avoids boundary conditions, or on $\mathbb{R}^d$ under sufficient moment and decay assumptions. We present smooth equations first and indicate where weak or viscosity solutions are needed. Systematic treatments include Cardaliaguet (2010); Bensoussan et al. (2013); Carmona and Delarue (2018a,b); Cardaliaguet et al. (2019); the numerical viewpoint is surveyed by Achdou and Laurière (2020).

2 From an N -player game to a population law

The finite game. Consider N exchangeable players. Player i has state $X_t^i \in \mathbb{R}^d$ and chooses a progressively measurable control $a_t^i \in \mathbb{R}^d$. In the simplest model,

$$dX_t^i = a_t^i dt + \sqrt{2\nu} dB_t^i, \quad \nu \geq 0, \quad (1)$$

where the Brownian motions are independent. The other players enter only through the empirical measure

$$\alpha_t^{N,-i} = \frac{1}{N-1} \sum_{j \neq i} \delta_{X_t^j}. \quad (2)$$

Given a running Lagrangian $L(x, a)$, a running interaction $F(x, \alpha)$, and a terminal interaction $G(x, \alpha)$, player i minimizes

$$J_i^N(a^1, \dots, a^N) = \mathbb{E} \left[\int_0^T (L(X_t^i, a_t^i) + F(X_t^i, \alpha_t^{N,-i})) dt + G(X_T^i, \alpha_T^{N,-i}) \right]. \quad (3)$$

Definition 2.1 (Nash equilibrium). A control profile $(a^{1,*}, \dots, a^{N,*})$ is a Nash equilibrium if no player can lower its own cost by changing only its control: for every i and every admissible a^i ,

$$J_i^N(a^{1,*}, \dots, a^{N,*}) \leq J_i^N(a^{1,*}, \dots, a^{i-1,*}, a^i, a^{i+1,*}, \dots, a^{N,*}).$$

It is an ε -Nash equilibrium if the right-hand side may improve the cost by at most ε .

The equilibrium is non-cooperative: the feedback of player i minimizes its own objective while the strategies of all other players are frozen. This is different from minimizing $\sum_i J_i^N$, which is a centralized control problem.

The mean field ansatz. When N is large and each player has weight $1/N$, one expects a typical agent to have a negligible effect on the population law. The empirical measure is replaced by a deterministic curve $(\alpha_t)_{0 \leq t \leq T}$, and the representative agent solves a standard control problem with this curve frozen. The optimal controlled state must then have law α_t . Symbolically,

population curve $\mapsto$ optimal feedback $\mapsto$ law of the optimally controlled state.

An MFG equilibrium is a fixed point of this map.

Under suitable regularity and uniqueness assumptions, the feedback obtained from the MFG system yields an ε_N -Nash equilibrium for the finite game, with $\varepsilon_N \rightarrow 0$. The probabilistic mechanism is propagation of chaos: finitely many tagged players become asymptotically independent with common law α_t . Precise open-loop and closed-loop results require different arguments; see [Carmona et al. \(2016\)](#); [Lacker \(2020\)](#); [Cardaliaguet et al. \(2019\)](#).

3 The representative-agent control problem

Fix for the moment a continuous curve $t \mapsto \alpha_t \in \mathcal{P}_2(\mathbb{R}^d)$. Starting from $X_t = x$, a representative agent chooses a_s in

$$dX_s = a_s ds + \sqrt{2\nu} dB_s, \quad s \in [t, T], \quad (4)$$

and minimizes

$$\mathbb{E} \left[\int_t^T (L(X_s, a_s) + F(X_s, \alpha_s)) ds + G(X_T, \alpha_T) \right]. \quad (5)$$

Its value is denoted by $u(t, x)$.

We use the Hamiltonian convention

$$H(x, p) = \sup_{a \in \mathbb{R}^d} \{-a \cdot p - L(x, a)\}. \quad (6)$$

If $L(x, \cdot)$ is strictly convex and smooth, the maximizing control is

$$a^*(x, p) = -\nabla_p H(x, p). \quad (7)$$

For $L(x, a) = |a|^2/2$, one has $H(x, p) = |p|^2/2$ and $a^* = -p$.

Proposition 3.1 (Hamilton–Jacobi–Bellman equation). *Assume temporarily that the value function is smooth. For a prescribed population curve $(\alpha_t)_t$, the value function of (5) satisfies*

$$-\partial_t u - \nu \Delta u + H(x, \nabla u) = F(x, \alpha_t), \quad u(T, x) = G(x, \alpha_T). \quad (8)$$

The optimal Markov feedback is $a_t^*(x) = -\nabla_p H(x, \nabla u(t, x))$.

Proof. The dynamic-programming principle over a short interval $[t, t+h]$ gives

$$u(t, x) = \inf_a \mathbb{E} \left[\int_t^{t+h} (L(X_s, a_s) + F(X_s, \alpha_s)) \, ds + u(t+h, X_{t+h}) \right].$$

Apply Itô's formula to $u(s, X_s)$, divide by h , and let $h \downarrow 0$. The local minimization is $\inf_a \{L(x, a) + a \cdot \nabla u\} = -H(x, \nabla u)$, which gives (8). The optimizer is (7). $\square$

When smoothness fails, the HJB equation is naturally interpreted in the viscosity sense. This is one reason why the backward equation and the forward equation do not live in identical function spaces.

4 The mean field game PDE system

Closing the fixed point. Apply the optimal feedback from Proposition 3.1 to an initial random state $X_0 \sim \alpha_0$. Its law solves the Kolmogorov forward equation. Requiring this law to equal the curve used in the control problem closes the system.

Definition 4.1 (Finite-horizon mean field game system). Given an initial law α_0 , a classical MFG equilibrium is a pair (u, α) satisfying

$$\begin{cases} -\partial_t u_t - \nu \Delta u_t + H(x, \nabla u_t) = F(x, \alpha_t), & u_T(x) = G(x, \alpha_T), \\ \partial_t \alpha_t - \nu \Delta \alpha_t - \operatorname{div}(\alpha_t \nabla_p H(x, \nabla u_t)) = 0, & \alpha_{t=0} = \alpha_0. \end{cases} \quad (9)$$

The second equation is understood distributionally when α_t has no density.

The first equation runs backward because its terminal value is prescribed; the second runs forward because its initial law is prescribed. Neither can be solved independently: F and G couple the population into the HJB equation, while u determines the drift in the Fokker–Planck equation.

Proposition 4.2 (Weak forward equation and mass conservation). *Let $b_t(x) = -\nabla_p H(x, \nabla u_t(x))$. The forward equation in (9) is equivalent to*

$$\frac{d}{dt} \int \varphi(x) \, d\alpha_t(x) = \int (\nabla \varphi(x) \cdot b_t(x) + \nu \Delta \varphi(x)) \, d\alpha_t(x) \quad (10)$$

for every smooth test function φ . In particular, $\alpha_t(\mathcal{X}) = \alpha_0(\mathcal{X})$.

Proof. Multiply the Fokker–Planck equation by φ and integrate by parts. Taking $\varphi \equiv 1$ gives mass conservation. $\square$

For $\nu = 0$, the forward equation is exactly the continuity equation

$$\partial_t \alpha_t + \operatorname{div}(\alpha_t v_t) = 0, \quad v_t = -\nabla_p H(x, \nabla u_t). \quad (11)$$

This is the same Eulerian kinematics used by dynamic optimal transport and Wasserstein gradient flows. For $\nu > 0$, it is a Fokker–Planck equation. If $\alpha_t = \rho_t \, dx$ is smooth and positive, it can also be rewritten as a continuity equation with current velocity

$$\partial_t \rho_t + \operatorname{div}(\rho_t \tilde{v}_t) = 0, \quad \tilde{v}_t = -\nabla_p H(x, \nabla u_t) - \nu \nabla \log \rho_t. \quad (12)$$

The score correction is the same one that relates Langevin diffusion and deterministic probability-flow representations.

Remark 4.3 (What is forward and what is backward). The density equation is causal once a feedback is known. The feedback is not causal in the population curve because a rational agent anticipates future congestion and the terminal cost. This anticipation is encoded by the backward HJB equation. Setting u_t from an instantaneous cost only would produce a myopic interacting-particle or gradient-flow model, not the same game.

5 An optimal-transport dictionary

The analogy with the rest of the book is clearest at the level of mathematical objects rather than slogans.

Optimal-transport/PDE object	Role in a mean field game
Probability measure α_t	Distribution of the state of a representative agent.
Empirical measure	Finite-player approximation of the population law.
Continuity equation	Deterministic transport of the population by the equilibrium feedback.
Fokker–Planck equation	Transport plus idiosyncratic Brownian exploration.
Kinetic action	Individual effort, and in potential games the dynamic transport term of the social potential.
Hamilton–Jacobi equation	Dual/optimality equation for a representative agent.
First variation	Converts a potential functional of α into the coupling cost perceived by one agent.
JKO step	A proximal, myopic evolution of measures; related to but distinct from a finite-horizon MFG equilibrium.
Geodesic convexity	A useful convexity mechanism for potential planning problems, not a replacement for Nash monotonicity.

The Benamou–Brenier limit. If $F = 0$, G imposes a hard terminal law, $\nu = 0$, and $L(a) = |a|^2/2$, the population planning problem reduces to dynamic quadratic optimal transport. The HJB equation becomes the Hamilton–Jacobi equation dual to the Benamou–Brenier formulation, and the optimal population velocity is a gradient. MFGs enlarge this picture by adding endogenous running and terminal interactions, while generic games lose the global variational structure entirely (Benamou and Brenier, 2000).

Schrödinger bridges. With $\nu > 0$ and prescribed endpoint laws, minimizing quadratic drift energy relative to Brownian motion gives the Schrödinger bridge. A second-order potential MFG has a similar controlled-diffusion constraint, but its final law is usually endogenous and its congestion or interaction potential is integrated along the path. The entropy-on-path-space representation of variational second-order MFGs is developed by Benamou et al. (2019).

6 Potential mean field games as variational problems

A generic MFG is only a fixed point. A particularly important subclass is generated by functionals on the space of measures.

Definition 6.1 (Potential couplings). The running and terminal couplings are potential if there exist functionals $\mathcal{F}, \mathcal{G} : \mathcal{P}(X) \rightarrow \mathbb{R} \cup \{+\infty\}$ such that

$$\frac{\delta \mathcal{F}}{\delta \alpha}(\alpha)(x) = F(x, \alpha), \quad \frac{\delta \mathcal{G}}{\delta \alpha}(\alpha)(x) = G(x, \alpha), \quad (13)$$

up to additive constants independent of x .

For clarity, first take $L(a) = |a|^2 / 2$. The potential planning problem is

$$\inf_{(\alpha, v)} \left\{ \int_0^T \left[\frac{1}{2} \int |v_t(x)|^2 d\alpha_t(x) + \mathcal{F}(\alpha_t) \right] dt + \mathcal{G}(\alpha_T) \right\}, \quad (14)$$

subject to

$$\partial_t \alpha_t + \operatorname{div}(\alpha_t v_t) = v \Delta \alpha_t, \quad \alpha_{t=0} = \alpha_0. \quad (15)$$

The unknown final law is selected by the terminal potential. Prescribing $\alpha_T = \alpha_1$ instead gives the mean field planning problem.

Proposition 6.2 (Optimality system of a quadratic potential MFG). *Assume all fields are smooth, $\alpha_t = \rho_t dx$ is positive, and (14) admits a minimizer. Then its first-order optimality conditions are*

$$\begin{cases} -\partial_t u - v \Delta u + \frac{1}{2} |\nabla u|^2 = F(x, \alpha_t), \\ \partial_t \alpha_t - v \Delta \alpha_t - \operatorname{div}(\alpha_t \nabla u) = 0, \\ \alpha_{t=0} = \alpha_0, \quad u_T = G(\cdot, \alpha_T). \end{cases} \quad (16)$$

Thus a minimizer gives an MFG equilibrium.

Proof. Introduce the multiplier $-u$ for (15), namely add

$$- \int_0^T \int u (\partial_t \rho + \operatorname{div}(\rho v) - v \Delta \rho) dx dt$$

to the objective. After integration by parts, the terms depending on v are $\int \rho(|v|^2 / 2 + v \cdot \nabla u)$. Their minimizer is $v = -\nabla u$. Variation in ρ , together with (13), gives the HJB equation. The free terminal variation gives $u_T = G(\cdot, \alpha_T)$. These are precisely (16). $\square$

The sign bookkeeping in the proof is conventional; the invariant statement is that minimizing the kinetic term produces the Legendre relation between the velocity and the HJB momentum. For a general convex Lagrangian, the Eulerian action is

$$\int_0^T \int L(x, v_t(x)) d\alpha_t(x) dt, \quad (17)$$

and the same calculation yields $v_t = -\nabla_p H(x, \nabla u_t)$.

Convex momentum variables. As in dynamic OT, write the momentum measure $m_t = \alpha_t v_t$. For a density ρ and momentum density m , the quadratic kinetic perspective is

$$J(\rho, m) = \begin{cases} \frac{|m|^2}{2\rho}, & \rho > 0, \\ 0, & (\rho, m) = (0, 0), \\ +\infty, & \rho = 0, m \neq 0. \end{cases} \quad (18)$$

It is jointly convex and positively one-homogeneous. Therefore (14) becomes a convex problem in (ρ, m) when $\mathcal{F}$ and $\mathcal{G}$ are convex, under the linear constraint

$$\partial_t \rho + \operatorname{div} m = v \Delta \rho. \quad (19)$$

This is the precise bridge to Benamou–Brenier algorithms and primal–dual splitting; see Benamou and Carlier (2015); Benamou et al. (2017).

Example 6.3 (Local congestion). Suppose $\alpha = \rho \, dx$ and agents pay $F(x, \alpha) = f(\rho(x))$, where f is increasing. Let $U'(r) = f(r)$. Then

$$\mathcal{F}(\alpha) = \int U(\rho(x)) \, dx$$

has first variation F . The convexity of U expresses aversion to congestion. A hard density cap is obtained by taking $U(r) = +\infty$ for $r > \kappa$, leading to a pressure multiplier.

Example 6.4 (Nonlocal interactions). Let

$$F(x, \alpha) = V(x) + \int W(x, y) \, d\alpha(y),$$

with a symmetric kernel $W(x, y) = W(y, x)$. Then

$$\mathcal{F}(\alpha) = \int V \, d\alpha + \frac{1}{2} \iint W(x, y) \, d\alpha(x) \, d\alpha(y).$$

Convexity of this functional is a positive-semidefiniteness condition on the kernel over signed measures of zero mass; it should not be confused with pointwise convexity of $W(x, y)$.

Remark 6.5 (Potential does not mean cooperative). The agents remain non-cooperative. Potentiality means that unilateral first-order incentives integrate to one scalar functional. Minimizing that functional recovers the Nash equilibrium, just as a finite-dimensional potential game can be solved by minimizing its potential. A generic social planner has different first-order conditions because changing the population law changes every agent's cost.

7 JKO schemes, planning, and gradient flows

The variational formula (14) looks close to a Wasserstein gradient flow, but the two constructions should not be identified.

The JKO construction. For an energy $f : \mathcal{P}_2(\mathcal{X}) \rightarrow \mathbb{R} \cup \{+\infty\}$, the JKO step with time step $\tau > 0$ is

$$\alpha^{k+1} \in \arg \min_{\alpha \in \mathcal{P}_2(\mathcal{X})} \left\{ \frac{1}{2\tau} \mathcal{W}_2^2(\alpha^k, \alpha) + f(\alpha) \right\}. \quad (20)$$

As $\tau \rightarrow 0$, its interpolation converges under standard assumptions to the Wasserstein gradient flow

$$\partial_t \alpha_t = \operatorname{div} \left(\alpha_t \nabla \frac{\delta f}{\delta \alpha}(\alpha_t) \right). \quad (21)$$

This evolution is initial-value and dissipative: the current measure determines the next proximal step, and $f(\alpha_t)$ decreases.

By contrast, a finite-horizon MFG is an equilibrium over an entire time interval. Its HJB equation is terminal-value and agents anticipate future population costs. Even a potential MFG minimizes one global action over the whole curve; it is closer to a problem in the calculus of variations than to a sequence of greedy descents.

Proposition 7.1 (A JKO step is a one-period planning problem). *The JKO problem (20) is equivalent to*

$$\inf_{(\alpha_s, v_s)} \left\{ \frac{1}{2\tau} \int_0^1 \int |v_s|^2 \, d\alpha_s \, ds + f(\alpha_1) \right\}, \quad (22)$$

subject to $\partial_s \alpha_s + \operatorname{div}(\alpha_s v_s) = 0$ and $\alpha_{s=0} = \alpha^k$, with α_1 free.

Proof. For each candidate endpoint α_1 , the Benamou–Brenier formula says that the infimum of the kinetic action over connecting curves is $\mathcal{W}_2^2(\alpha^k, \alpha_1)$. Minimizing also over α_1 gives (20). $\square$

The minimizing-movement construction is due to [Jordan et al. \(1998\)](#); its metric-space theory is developed by [Ambrosio et al. \(2008\)](#). Proposition 7.1 is the dynamic-transport reading of the same step.

This proposition gives a useful interpretation: a JKO step is a short-horizon planning problem with terminal potential f . Repeating the problem after each short interval produces a receding-horizon or myopic dynamics. A finite-horizon MFG instead solves one rational-expectations fixed point over $[0, T]$.

Remark 7.2 (Three variational objects). It is useful to keep three levels separate.

1. The Benamou–Brenier problem fixes both endpoint measures and minimizes kinetic action.
2. A potential MFG fixes the initial measure, usually leaves the terminal measure free, and adds running and terminal population potentials.
3. A JKO step is a single short planning problem whose terminal potential is the energy to be decreased; iteration and the limit $\tau \rightarrow 0$ create a gradient flow.

The formulas overlap, but their boundary conditions and time semantics do not.

Mean field control versus mean field games. Suppose a planner minimizes

$$\mathbb{E} \left[\int_0^T (L(X_t, a_t) + F(X_t, \operatorname{Law}(X_t))) \, dt + G(X_T, \operatorname{Law}(X_T)) \right]. \quad (23)$$

The planner internalizes how changing the law modifies the costs of all agents. Its optimality equation therefore contains derivatives of F and G with respect to the measure argument. In the MFG best response, one agent treats $(\alpha_t)_t$ as fixed and does not differentiate through it. The two systems coincide only under special potential structures, and even then the scalar potential minimized by the game must be distinguished from the sum of individual costs.

8 Monotonicity, convexity, and uniqueness

Potentiality is sufficient for a global minimization principle, but it is not necessary for well-posedness. The central game-theoretic condition introduced by Lasry and Lions is monotonicity in the measure variable.

Existence as a fixed point. The MFG construction itself suggests an existence proof: freeze a population curve, solve the backward control problem, propagate the resulting feedback forward, and look for a fixed point. The following intentionally strong version isolates this mechanism without entering the optimal regularity theory.

Theorem 8.1 (A basic parabolic existence principle). *Let $\mathcal{X} = \mathbb{T}^d$ and $v > 0$. Assume that H is smooth and uniformly convex in its momentum variable, with at most quadratic growth. Assume that $F(\cdot, \alpha)$ and $G(\cdot, \alpha)$ are uniformly bounded in $C^2(\mathbb{T}^d)$ and depend continuously on α for the narrow topology. If α_0 has a smooth positive density, then the system (9) admits at least one classical solution.*

Proof. Fix a continuous measure curve $\beta = (\beta_t)_{0 \leq t \leq T}$. Parabolic HJB theory gives a solution u^β with estimates controlled uniformly by the assumed bounds on F and G . Hence the feedback $b^\beta = -\nabla_p H(\cdot, \nabla u^\beta)$ is bounded and Hölder regular. Solving the linear Fokker–Planck equation with drift b^β and initial law α_0 defines a new curve $\Phi(\beta)$.

Parabolic compactness makes Φ map a closed convex bounded set of Hölder-continuous density curves into a relatively compact subset of itself. Stability of the HJB and Fokker–Planck equations with respect to their coefficients makes Φ continuous. Schauder’s fixed-point theorem gives a curve $\alpha = \Phi(\alpha)$. Together with u^α , it solves (9). The diffusion and smooth nonlocal couplings are what make this short classical argument work; local or singular couplings require weak-solution and compactness techniques. $\square$

This fixed-point route is conceptually distinct from the direct method for a potential game. In the potential case, coercivity and lower semicontinuity can produce a minimizer of (14); the PDE system is then recovered from its optimality conditions. General existence theories and substantially weaker assumptions are developed in Lasry and Lions (2007); Cardaliaguet (2010).

Definition 8.2 (Lasry–Lions monotonicity). A coupling $F : \mathcal{X} \times \mathcal{P}(\mathcal{X}) \rightarrow \mathbb{R}$ is monotone if

$$\int_{\mathcal{X}} (F(x, \alpha) - F(x, \beta)) \, d(\alpha - \beta)(x) \geq 0 \quad (24)$$

for all α, β . It is strictly monotone if equality forces $\alpha = \beta$ in the relevant class. The same definition applies to G .

Local congestion $F(x, \rho \, dx) = f(\rho(x))$ is monotone whenever f is increasing. A nonlocal linear coupling $F = W * \alpha$ is monotone exactly when $\iint W(x, y) \, d\eta(x) \, d\eta(y) \geq 0$ for every signed zero-mass measure η . When $F = \delta \mathcal{F} / \delta \alpha$, condition (24) is the monotonicity of the first variation and is implied by ordinary convexity of $\mathcal{F}$ under linear mixtures.

Theorem 8.3 (Uniqueness under monotonicity). *Let $H(x, \cdot)$ be strictly convex, and suppose F and G satisfy Lasry–Lions monotonicity. Assume two smooth solutions (u, α) and $(\tilde{u}, \tilde{\alpha})$ of (9) have the same initial law. Then their equilibrium feedbacks agree $\alpha_t \, dt$ - and $\tilde{\alpha}_t \, dt$ -almost everywhere. If one coupling is strictly monotone, the population curves agree; after a normalization, so do the value functions.*

Proof. Write $w = u - \tilde{u}$ and $\eta = \alpha - \tilde{\alpha}$. Differentiate the dual pairing $I(t) = \int w_t \, d\eta_t$, use the two HJB equations and the two forward equations, and integrate by parts. The diffusion terms cancel. Integrating from 0 to T , using $\eta_0 = 0$ and the terminal conditions, gives

$$\begin{aligned} & \int (G(x, \alpha_T) - G(x, \tilde{\alpha}_T)) \, d(\alpha_T - \tilde{\alpha}_T)(x) \\ & + \int_0^T \int (F(x, \alpha_t) - F(x, \tilde{\alpha}_t)) \, d(\alpha_t - \tilde{\alpha}_t)(x) \, dt \\ & + \int_0^T \int D_H(\nabla u_t, \nabla \tilde{u}_t) \, d\alpha_t \, dt + \int_0^T \int D_H(\nabla \tilde{u}_t, \nabla u_t) \, d\tilde{\alpha}_t \, dt = 0, \end{aligned} \quad (25)$$

where

$$D_H(p, q) = H(x, q) - H(x, p) - \nabla_p H(x, p) \cdot (q - p) \geq 0$$

is the Bregman divergence of the convex Hamiltonian. Every term in (25) is nonnegative. Hence all vanish. Strict convexity of H identifies the gradients, and strict monotonicity identifies the measures. The PDEs then identify the remaining quantities up to the usual additive constant in u when applicable. $\square$

The calculation is the MFG counterpart of a monotone-operator energy estimate. It is not the contraction estimate of a Wasserstein gradient flow, although both exploit a positive pairing between a difference of states and a difference of forces.

Two notions of convexity. Lasry–Lions monotonicity concerns linear interpolation of measures and the pairing of first variations. Wasserstein geodesic convexity concerns displacement interpolations. Neither notion implies the other in complete generality. For a potential game, linear convexity is the natural condition for convexity of the Eulerian momentum formulation, while displacement convexity can provide additional estimates along transport geodesics. The distinction is important: importing a convergence theorem from Wasserstein gradient flows into an MFG requires checking that the dynamics is actually a gradient flow, not merely that the state variable is a probability measure.

9 Canonical examples

9.1 Crowd motion with congestion

Take $L(a) = |a|^2/2$, a desired-location potential V , and a local congestion cost $f(\rho)$. The first-order system is

$$\begin{cases} -\partial_t u + \frac{1}{2} |\nabla u|^2 = V(x) + f(\rho), \\ \partial_t \rho - \operatorname{div}(\rho \nabla u) = 0. \end{cases} \quad (26)$$

The density anticipates bottlenecks because u is solved backward. This is different from a purely repulsive McKean–Vlasov flow, where the drift reacts only to the current density. Increasing f yields Lasry–Lions monotonicity; if $f = U'$, the game is potential with internal energy $\int U(\rho)$.

9.2 Quadratic and linear–quadratic games

Linear dynamics, quadratic effort, and couplings through the population mean lead to Riccati equations. For example, let

$$F(x, \alpha) = \frac{q}{2} |x - \bar{x}_\alpha|^2, \quad \bar{x}_\alpha = \int y \, d\alpha(y). \quad (27)$$

A quadratic ansatz for u closes the HJB equation, while the forward equation closes on the mean and covariance. These models are the most direct test bed for comparing finite-player Nash systems, MFG limits, and mean field control; see [Huang et al. \(2006\)](#); [Bensoussan et al. \(2013\)](#).

The coupling (27) should be interpreted carefully: agents are attracted toward the current mean when $q > 0$. Depending on signs and terminal terms, related models can instead penalize crowding or encourage dispersion.

9.3 Multi-population games

For K populations, the state is a vector of measures $\alpha = (\alpha^1, \dots, \alpha^K)$. Each population has its own value function and forward equation,

$$\begin{cases} -\partial_t u^k - v_k \Delta u^k + H_k(x, \nabla u^k) = F_k(x, \alpha_t), \\ \partial_t \alpha_t^k - v_k \Delta \alpha_t^k - \operatorname{div}(\alpha_t^k \nabla_p H_k(x, \nabla u_t^k)) = 0. \end{cases} \quad (28)$$

Cross-couplings can encode attraction, segregation, competition for a shared resource, or heterogeneous mobility. Potentiality requires the vector field of couplings $(F_1, \dots, F_K)$ to be the first variation of one functional of all populations, an infinite-dimensional analogue of symmetry of cross derivatives.

9.4 Stationary and ergodic games

Over a long horizon, one often seeks a stationary density α , a value function u , and an ergodic constant λ :

$$\begin{cases} -v\Delta u + H(x, \nabla u) = F(x, \alpha) + \lambda, \\ -v\Delta \alpha - \operatorname{div}(\alpha \nabla_p H(x, \nabla u)) = 0, \\ \alpha(X) = 1. \end{cases} \quad (29)$$

The scalar λ is the equilibrium average cost per unit time. It plays the role of an eigenvalue in the HJB equation. Long-time convergence of finite-horizon systems toward ergodic equilibria is subtle because the system is forward–backward; see [Cardaliaguet and Porretta \(2019\)](#).

10 The master equation: a PDE on Wasserstein space

The pair (u_t, α_t) solves one MFG for one initial distribution. To encode all initial laws simultaneously, introduce

$$U(t, x, \alpha), \quad (30)$$

the value at position x , time t , and population law α . Along an equilibrium curve,

$$u(t, x) = U(t, x, \alpha_t). \quad (31)$$

This is analogous to solving a finite-dimensional Hamilton–Jacobi equation and then evaluating it along characteristics, except that the state variable now contains a probability measure.

Write $\delta U / \delta \alpha(t, x, \alpha)(y)$ for the linear first variation in the measure and

$$D_\alpha U(t, x, \alpha)(y) = \nabla_y \frac{\delta U}{\delta \alpha}(t, x, \alpha)(y) \quad (32)$$

for its intrinsic, or Lions, derivative. In the absence of common noise, the formal master equation associated with (9) is

$$\begin{aligned} -\partial_t U - v\Delta_x U + H(x, \nabla_x U) - v \int \operatorname{div}_y D_\alpha U(t, x, \alpha)(y) \, d\alpha(y) \\ + \int D_\alpha U(t, x, \alpha)(y) \cdot \nabla_p H(y, \nabla_x U(t, y, \alpha)) \, d\alpha(y) = F(x, \alpha), \end{aligned} \quad (33)$$

with terminal condition $U(T, x, \alpha) = G(x, \alpha)$.

Proposition 10.1 (The MFG system as characteristics of the master equation). *Assume U is a smooth solution of (33). Let α_t solve the forward equation driven by $u_t(x) = U(t, x, \alpha_t)$. Then (u, α) solves the MFG system (9).*

Proof. Differentiate (31) along α_t :

$$\partial_t u(t, x) = \partial_t U(t, x, \alpha_t) + \int \frac{\delta U}{\delta \alpha}(t, x, \alpha_t)(y) \, \partial_t \alpha_t(dy).$$

Insert the forward Fokker–Planck equation and integrate by parts in y . The diffusion gives $v \int \operatorname{div}_y D_\alpha U \, d\alpha_t$, and the drift gives the integral containing $D_\alpha U \cdot \nabla_p H$, with the signs displayed in (33). The master equation therefore reduces to the HJB equation along the curve. The forward equation holds by construction. $\square$

Common noise makes the population law itself stochastic and adds second-order derivatives with respect to the measure variable. The master equation is also the key object for proving that the N -player Nash system converges to the MFG limit; see [Cardaliaguet et al. \(2019\)](#).

Remark 10.2 (Measure derivatives versus Wasserstein gradients). The scalar first variation $\delta U/\delta\alpha$ and the intrinsic derivative $D_\alpha U = \nabla_y \delta U/\delta\alpha$ mirror the book's distinction between a functional derivative and its Wasserstein gradient. The master equation is not itself a Wasserstein gradient-flow equation: it is a Hamilton–Jacobi equation on a space whose state includes a measure.

11 Numerical viewpoints

The forward–backward structure is the main numerical difficulty. A naive time march cannot solve both boundary conditions simultaneously.

11.1 Forward–backward fixed-point iteration

The simplest method alternates best response and population propagation. Starting from a guessed curve $\alpha^{(0)}$, repeat

1. solve the HJB equation backward using $\alpha^{(k)}$;
2. propagate the Fokker–Planck equation forward using the resulting feedback, obtaining $\hat{\alpha}^{(k+1)}$;
3. damp the update, $\alpha^{(k+1)} = (1 - \theta_k)\alpha^{(k)} + \theta_k\hat{\alpha}^{(k+1)}$.

This is a best-response or fictitious-play mechanism. Convergence is not automatic, but monotonicity and suitable averaging can stabilize it; see [Hadikhanloo \(2017\)](#); [Cardaliaguet and Hadikhanloo \(2017\)](#).

Forward–backward MFG iteration

Input: initial law α_0 , horizon T , couplings F, G , Hamiltonian H , damping sequence $(\theta_k)_k$.

Initialize: choose a population path $\alpha^{(0)} = (\alpha_t^{(0)})_{0 \leq t \leq T}$.

For $k = 0, 1, \dots$

$u_T^{(k)} \leftarrow G(\cdot, \alpha_T^{(k)})$.

Solve backward: $-\partial_t u^{(k)} - \nu \Delta u^{(k)} + H(x, \nabla u^{(k)}) = F(x, \alpha_t^{(k)})$.

Solve forward: $\partial_t \hat{\alpha}^{(k+1)} - \nu \Delta \hat{\alpha}^{(k+1)} - \operatorname{div}(\hat{\alpha}^{(k+1)} \nabla_p H(x, \nabla u^{(k)})) = 0$, $\hat{\alpha}_{t=0}^{(k+1)} = \alpha_0$.

$\alpha^{(k+1)} \leftarrow (1 - \theta_k)\alpha^{(k)} + \theta_k\hat{\alpha}^{(k+1)}$.

Stop if the path residual is below tolerance.

Output: approximate equilibrium $(u^{(k)}, \alpha^{(k)})$.

11.2 Variational and primal–dual algorithms

For potential games, the momentum formulation (18)–(19) is often preferable. It exposes a convex objective under a linear space–time PDE constraint. Augmented-Lagrangian, Douglas–Rachford, ADMM, and primal–dual methods then alternate a local proximal step for the perspective/congestion integrand with a global projection onto the discrete continuity or Fokker–Planck equation. This is exactly the algorithmic architecture used for dynamic optimal transport, with additional running and terminal potentials ([Benamou and Carlier, 2015](#); [Benamou et al., 2017](#)).

With diffusion and quadratic Hamiltonian, a path-space entropy formulation can turn the kinetic control cost into relative entropy with respect to Brownian motion. Time discretization then yields multi-marginal entropic problems and Sinkhorn-like scaling algorithms (Benamou et al., 2019).

11.3 Finite differences, particles, and learning

Monotone finite-difference schemes are natural for the HJB equation, while mass-conservative schemes are required for the Fokker–Planck equation. Discrete adjointness between the two is valuable: it preserves the duality used in uniqueness and variational arguments. Convergence of representative schemes is developed in Achdou et al. (2013); see Achdou and Laurière (2020) for a broader account.

Particle methods approximate α_t by an empirical measure. They are especially natural for probabilistic formulations and high-dimensional problems, but estimating density-dependent congestion from particles requires regularization. Neural solvers can parametrize u , the feedback, or the master field $U(t, x, \alpha)$; doing so does not remove the equilibrium constraint, and training residuals must still control both the backward and forward equations.

12 What to remember

1. An MFG is the limit of a large symmetric non-cooperative game. Its state is a population law, but its defining principle is Nash consistency.
2. The value function solves a backward HJB equation; the population law solves a forward continuity or Fokker–Planck equation. Their coupling is a fixed point between optimal response and induced law.
3. Optimal transport supplies the kinematics, kinetic action, momentum perspective, and Hamilton–Jacobi duality. Potential MFGs become Benamou–Brenier-type variational problems over curves of measures.
4. A generic MFG is not a Wasserstein gradient flow. JKO is best viewed here as a one-step/myopic planning construction; repeating short planning problems creates a dissipative flow, whereas an MFG encodes anticipation over a fixed horizon.
5. Lasry–Lions monotonicity is the central uniqueness mechanism. It is related to convexity of potential couplings but distinct from displacement convexity along Wasserstein geodesics.
6. The master equation is a Hamilton–Jacobi equation on the space of positions and probability measures. It organizes dependence on the initial population and the convergence from finite-player Nash systems.

The conceptual payoff is a unified but non-reductive picture. Optimal transport, Wasserstein flows, mean field control, and mean field games share the language of evolving probability measures. They differ in who chooses the velocity, which endpoint data are prescribed, and whether the governing principle is transport optimality, dissipation, centralized control, or Nash equilibrium. Keeping these distinctions visible is what makes the common PDE and variational machinery useful.

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